

Prim's Algorithm — DAA

Prim's algorithm is a greedy algorithm used to find the Minimum Spanning Tree (MST) of a weighted, connected, undirected graph. This version uses a simple adjacency matrix.

Algorithm

1. Start with any vertex.
2. Mark the starting vertex as visited.
3. Find the smallest edge connecting a visited vertex to an unvisited vertex.
4. Add that edge and the new vertex to the MST.
5. Repeat until $V - 1$ edges are selected.
6. Add all selected edge weights to get the minimum cost.

C Code

```
#include <stdio.h>

int main()
{
    int graph[5][5] = {
        {0,2,0,6,0},
        {2,0,3,8,5},
        {0,3,0,0,7},
        {6,8,0,0,9},
        {0,5,7,9,0}
    };

    int visited[5] = {1,0,0,0,0};
    int edges = 0, min, x, y, cost = 0;

    while (edges < 4)
    {
        min = 999;

        for (int i = 0; i < 5; i++)
        {
            if (visited[i])
            {
                for (int j = 0; j < 5; j++)
                {
                    if (!visited[j] && graph[i][j] != 0)
                    {
                        if (graph[i][j] < min)
                        {
                            min = graph[i][j];
                            x = i;
                            y = j;
                        }
                    }
                }
            }
        }

        printf("%d - %d = %d\n", x, y, min);
        visited[y] = 1;
        cost += min;
        edges++;
    }

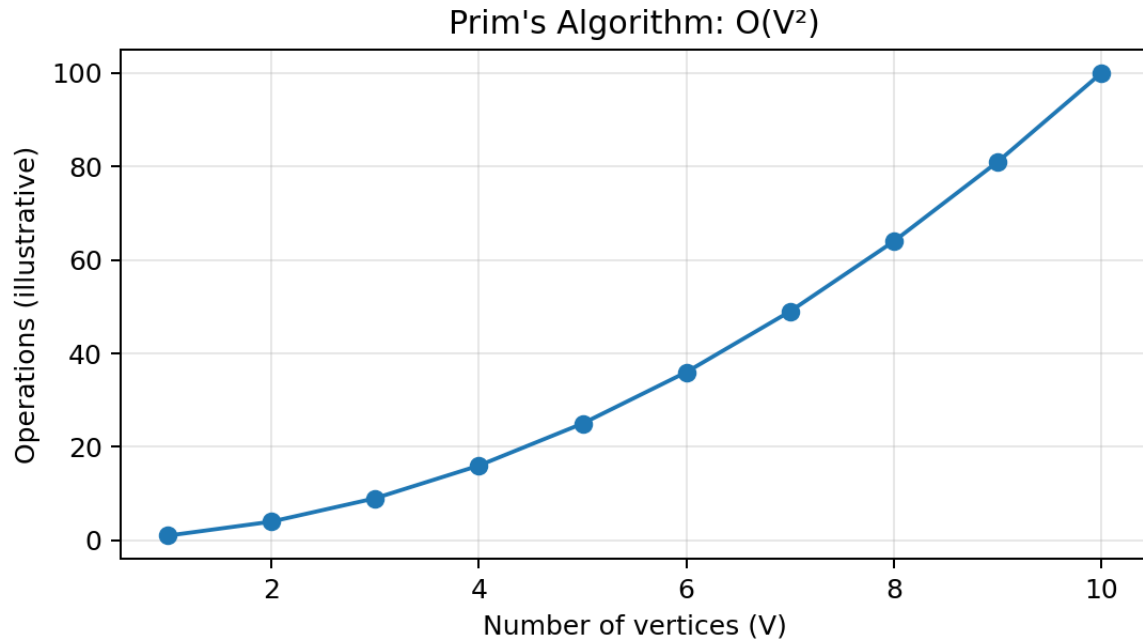
    printf("Minimum Cost = %d", cost);
    return 0;
}
```

}

Time Complexity

Time Complexity: $O(V^2)$

For this adjacency-matrix implementation, the work grows approximately as the square of the number of vertices.



Example Output

```
0 - 1 = 2
1 - 2 = 3
0 - 3 = 6
1 - 4 = 5
Minimum Cost = 16
```

Viva / Exam Point

Prim's is a **greedy algorithm**. It repeatedly chooses the smallest edge from the visited set to the unvisited set. An MST contains $V - 1$ edges.